MODULE *problem1*

EXTENDS *TLC*, *Integers*

VARIABLES *interested*, *pc*, *turn*

$processes \triangleq \{0, 1\}$

$TypeOK \triangleq$
$\quad \land \quad interested \in [processes \to \text{BOOLEAN}\ ]$
$\quad \land \quad turn \in processes$
$\quad \land \quad pc \in [processes \to \{\text{“T1”}, \text{“T2”}, \text{“T3A”}, \text{“T3B”}, \text{“E1”}\}]$

$Init \triangleq$
$\quad \land\, interested = [i \in processes \mapsto \text{FALSE}]$
$\quad \land\, turn \in processes$
$\quad \land\, pc = [p \in processes \mapsto \text{“T1”}]$

$T1(p) \triangleq$
$\quad \land\, pc[p] = \text{“T1”}$
$\quad \land\, pc' = [pc\ \text{EXCEPT}\ ![p] = \text{“T2”}]$
$\quad \land\, interested' = [interested\ \text{EXCEPT}\ ![p] = \text{TRUE}]$
$\quad \land\, \text{UNCHANGED}\ turn$

$T2(p) \triangleq$
$\quad \land\, pc[p] = \text{“T2”}$
$\quad \land\, pc' = [pc\ \text{EXCEPT}\ ![p] = \text{“T3A”}]$
$\quad \land\, turn' = 1 - p$
$\quad \land\, \text{UNCHANGED}\ interested$

$$
\begin{array}{l}
T3A(p) \triangleq \\
\quad \land\ pc[p] = \text{“T3A”} \\
\quad \land\ pc' = [pc \text{ EXCEPT } ![p] = \\
\qquad \text{IF } turn = p \text{ THEN “E1”} \\
\qquad \text{ELSE “T3B”}] \\
\quad \land \text{ UNCHANGED } \langle turn, interested \rangle
\end{array}
$$

$$
\begin{array}{l}
T3B(p) \triangleq \\
\quad \land\ pc[p] = \text{“T3B”} \\
\quad \land\ pc' = [pc \text{ EXCEPT } ![p] = \\
\qquad \text{IF } interested[1-p] \text{ THEN “T3A”} \\
\qquad \text{ELSE “E1”}] \\
\quad \land \text{ UNCHANGED } \langle turn, interested \rangle
\end{array}
$$

$$
\begin{array}{l}
E1(p) \triangleq \\
\quad \land\ pc[p] = \text{“E1”} \\
\quad \land\ pc' = [pc \text{ EXCEPT } ![p] = \text{“T1”}] \\
\quad \land\ interested' = [interested \text{ EXCEPT } ![p] = \text{FALSE}] \\
\quad \land \text{ UNCHANGED } turn
\end{array}
$$

$$
\begin{array}{l}
Mutex \triangleq \\
\quad \neg(pc[1] = \text{“E1”} \land pc[0] = \text{“E1”})
\end{array}
$$

$$
\begin{array}{l}
Next \triangleq \\
\quad \exists\, p \in processes : \\
\qquad \lor\ T1(p) \\
\qquad \lor\ T2(p) \\
\qquad \lor\ T3A(p) \\
\qquad \lor\ T3B(p) \\
\qquad \lor\ E1(p)
\end{array}
$$